

Ultrafast X-Ray Experiments

**Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21**

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1. Einleitung

2. Theoretical Background

2.1. Synchrotronradiation

When accelerating electrons or other charged particles to relativistic speeds along a curved trajectory they emit so-called synchrotron radiation. It gains its name from the ring accelerators, the synchrotrons, where it was first observed. A typical synchrotron facility consists of a linear accelerator leading into a booster ring in which the electrons are brought up to relativistic speeds [1][2]. The electrons are then fed into a storage ring in which they continue to circulate at relativistic speeds. From the storage ring the beam-lines lead tangentially away in which the synchrotron radiation is utilized in experimental hutches [1][2]. Synchrotron radiation is bright X-radiation with a wide tunability in energy, which is useful for examining structures and dynamics of materials, as x-ray photons have energies in the order of several kiloelectronvolts (keV) [2]. This means their wavelengths are in the order of magnitude of Angstroms and comparable to the spacing of atoms in solid materials. This is useful, as it means that atoms in a crystalline array can diffract x-ray beams and the interference patterns can be measured [1].

There are four generations of synchrotrons with the main difference being the brilliance achieved [1]. The first was the bending magnet in the synchrotron as it was used for high energy collider physics with no additional modifications. The second generation introduced the wiggler, a configuration of dipole magnets with alternating polarity to achieve synchrotron radiation on a linear path. The third generation synchrotron radiation facilities were designed for low emittance of the beams with more effective linear magnet configurations, the undulators [2]. A low emittance means a smaller angular divergence of the beam and a higher brilliance [2].

2.1.1. Free Electron Laser (FEL)

Free-electron lasers are coherent radiation sources that are capable of even higher brilliance than third generation synchrotron facilities [3]. They are in theory able to operate at any arbitrary wavelength or spectrum of wavelength with the limit being the energy and quality of the accelerated electron beam [3]. The duration of a single X-ray pulse can also be reduced to even less than a few femtoseconds. This allows for an investigation of molecular processes and photochemical reactions through a form of sequential snapshots with a characteristic time of under a picosecond [1].

They use a process called self-amplified stimulated emission (SASE). SASE free-electron lasers typically use monochromatic electron beams with a long, high-quality undulator around the order of magnitude of a few hundred metres. The light waves emitted by the electron bunches will line up in phase and amplify themselves through superposition. As the electrons oscillate down the undulator they are also interacting with the magnetic components of the light they themselves are generating. The transverse component of the electrons oscillation and the magnetic field component of the emitted light create a Lorentz force which acts upon the electrons and moves them along axial directions. This leads to the electrons creating microbunches inside their conventional bunches [1].

2.1.2. Undulators

2.2. Pump-Probe-Principle

The principle of pump-probe-spectroscopy is sending two lasers at the same object with a certain time delay so that the first laser, an optical laser, can excite the system in some way and the second laser, an X-ray pulse can measure the response. In a FEL such as the European XFEL this can be done with ultrafast X-ray pulses on a timescale of a few femtoseconds (10^{-15} s). Varying the time delay allows for a series of sequential snapshots of the process observed and varying the wavelength of the excitation laser allows for an investigation of different processes [4].

2.2.1. The Pump-Pulse

When absorbing the energy of a photon, the energy of an atom increases as an electron is promoted into an excited energy level.

2.2.2. The Probe-Pulse

2.3. The investigated Molecule

2.3.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$

2.3.2. Photophysics

3. Experimental Setup

The entire experiment is conducted in VLab, a software modeled after a hutch in the experimental hall of the European XFEL in Schenefeld. The European XFEL is 3.4 km long and runs from the DESY research center to the European XFEL campus. A 1.9 km long superconducting linear accelerator feeds electrons with energies up to 17.5 GeV into a branched tunnel system with two 212 m long undulators that generate hard X-radiation in the range of 5 keV to 25 keV and one 125 m long undulator that generates soft X-radiation in the range of 0.25 keV to 3 keV.

VLab recreates the Femtosecond X-ray Experiment research platform.

Datenstruktur (beam imaging and x-ray eye): 420x420 pixel, bei biu screen dim von 10mm x 10mm, bei x-ray eye screen 2mm x 2mm

For further elaboration see Appendix B.

3.1. Undulator and Beamline

3.2. Monochromator and Focusing Optics

monochromator focusing optics: mirrors, power slits, solid attenuators, refractive lenses, X-ray eye

3.3. Spectrum Analyser

diffraction gratings

3.4. Time-arrival Detector

3.5. Sample Environment

3.6. Laser System

3.7. Von Hamos Spectrometer

Quelle Versuchsmappe:[4]

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.

As the absorption edge for the Fe $K\beta_{1,3}$ line of 7065 eV, the energy of the x-ray beam is set to 7400 eV, to avoid the ripples near the reaction (as illustrated by the X-ray absorption spectrum of $[Fe(terpy)_2]^{2+}$ Figure 1) but without too strong of a loss in intensity.

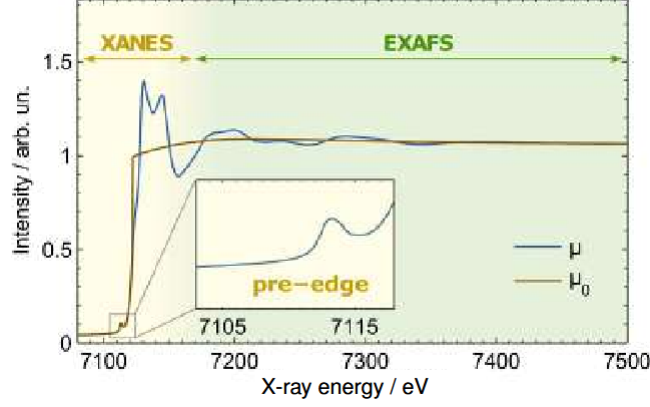


Figure 1: X-Ray absorption spectrum of $[\text{Fe}(\text{terpy})_2]^{2+}$ (terpy: terpyridin) at the Fe K edge displaying the XANES and EXAFS regions including the background absorption contributions μ_0 . The inset zooms into the weak pre edge features [4].

To measure the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturation of the beam imaging unit 2 (BIU2):

Table 1: Attenuator foils used

Rod	Attenuator foil
1	C 0.1
2	—
3	Si 0.1
4	B4C 0.8

The size and position of the x-ray is now measured using BIU2. Now the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24° .

For the following experiment the monochromatic beam is used as it leads to a higher resolution than the pink x-ray, even though the intensity of the beam suffers.

The last step of diagnosing the beam line is using the refractive beryllium x-ray lenses to focus the beam at the imaging unit x-ray eye. It's important to appropriately adjust the attentuator foils as to not burn the imaging unit with a focussed beam.

With beryllium lenses 1, 2 and 5 (taken from ??) and the attentuator foils listed here ??, a beam with the FWHM of 13 μm (siehe ??) is achieved. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [?] and the properties of the lenses themselves taken from ??:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$

$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4.76 \text{ m}$$

—Notes—

X-ray energy $E_0 = 7400 \text{ eV}$ (has to be a good bit away from reaction edge of 7065 keV to be stable)

Alignment: attenuators to not burn through out screens (here: 4: B4C 0.8, 3: C 0.4, 2: Si 0.05, C 0.1) size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

optical resolution: 25 $\mu\text{m}/\text{px}$

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

The transition metal complex $[\text{Fe}(\text{bipy})_3]^{2+}$ to be studied is obtained through dissolving the salt $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ in deionized water to the desired concentration. As the actual sample is a powder consisting of small microcrystals that contain incorporated water which remained in the crystal lattice after the crystallization process.

A minimum volume of 50 ml is required to operate the liquid jet in the experiment. To ensure that the sample is completely refreshed between probe pulses, the jet needs to have a certain flow speed with an assumed repetition rate of the X-ray pulse train of 1.1 MHz and an assumed top hat beam profile of the X-ray laser with the dimension $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$. The minimum required flowspeed is calculated as follows:

$$\Delta x_{\text{travel}} \geq w \Rightarrow v_{\text{min}} \cdot \Delta\tau \geq w \Rightarrow v_{\text{min}} \geq \frac{w}{\Delta\tau} \Rightarrow v_{\text{min}} = 11\text{ m s}^{-1}$$

With $w = 10\text{ }\mu\text{m}$ being the spatial width and $\Delta\tau = \frac{1}{f} = \frac{1}{1.1 \times 10^6\text{ Hz}} \approx 9.09 \times 10^{-7}\text{ s}$ the characteristic time.

There are three laser wavelengths available for excitation: 355 nm, 400 nm and 515 nm. To evaluate which fits the experiment best, the concentration c of the solution for the three wavelengths for a desired optical density of $OD = 1$ and a liquid jet thickness of $d = 25\text{ }\mu\text{m}$. The optical density is defined as $OD = \epsilon \cdot c \cdot d$ with $\epsilon(\lambda)$ the wavelength dependent extinction coefficient of $[\text{Fe}(\text{bipy})_3]^{2+}$ in water. The extinction coefficients can be estimated to be $\epsilon(355\text{ nm}) = 12\,000\text{ M}^{-1}\text{ cm}^{-1}$, $\epsilon(400\text{ nm}) = 4000\text{ M}^{-1}\text{ cm}^{-1}$, $\epsilon(515\text{ nm}) = 8000\text{ M}^{-1}\text{ cm}^{-1}$ as taken from Figure 2.

The following concentrations were calculated:

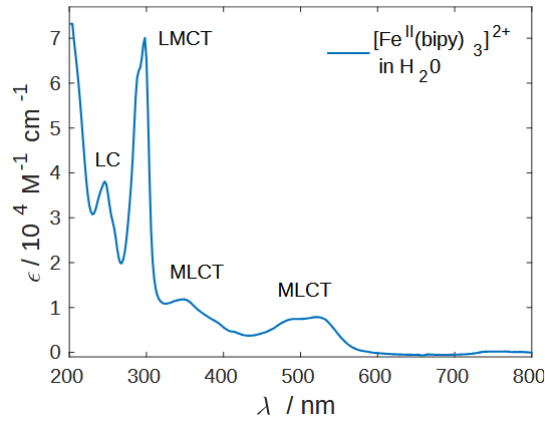


Figure 2: UV/VIS absorption spectrum of an aqueous $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ solution. The different CT transitions are indicated. The measured molar absorptivity ϵ values lie in the expected range of $10^{-4}\text{ l mol}^{-1}\text{ cm}^{-1}$ to $10^{-5}\text{ l mol}^{-1}\text{ cm}^{-1}$ for dipole-allowed transitions[4]

$$c_{355} = 0.033\text{ mol/L}, c_{400} = 0.100\text{ mol/L}, c_{515} = 0.050\text{ mol/L}$$

There are different pulse energies available for each excitation wavelength: $5\text{ }\mu\text{J}$ (355 nm), $15\text{ }\mu\text{J}$ (400 nm) and $50\text{ }\mu\text{J}$ (515 nm). With that in mind the wavelength of 400 nm was chosen, as it has a higher pulse energy, even though it's not ideal when looking at the low absorption rate. The extinction coefficient of the 515 nm is slightly better but with the low pulse energy it will be hard to see anything.

As we want to ensure that the X-ray beam always hits excited molecules we need to determine the diameter of the optical laser beam as well as the diameter of the X-ray beam correctly in relation to each other. If the optical laser and the X-ray laser had the same diameter, the x-ray would also investigate molecules that were excited only weakly or even not at all, as the optical laser has an approximately Gaussian beam shape. Ideally, the X-ray is positioned exactly in the centre of the ca. 10 times bigger

optical laser beam. Then it can investigate a maximal volume of excited molecules. This also compensates for the fact that the optical laser jitters in the focus layer of the X-ray due to not being focussed there itself.

These estimates have consequences for our needed flow speed of the liquid jet which has to be 10 times faster to adjust to the new diameter of the optical laser.

Ideally, the number of absorbed optical laser photons should be less than or approximately equal to the number of sample molecules in the laser-excited volume. This reduces the likelihood that multiple photons hit the same molecule and excite it into states other than the one we want to observe while still optimizing the number of excited molecules.

There are two X-ray crystal spectrometers available to use: a von Hamos spectrometer and a Johann spectrometer. As the von Hamos spectrometer can record the entire spectrum at once while also being immune to intensity variation, in contrast to the Johann spectrometer which needs to scan each wavelength separately and requires normalization to the accumulated incident intensity at each measurement point to measure the whole spectrum, it is better suited for this experiment.

In order to choose the correct crystal for the emission spectrometer the Bragg reflection angle needs to be considered.

4.2. Performing the Experiment with Data Acquisition

Using the Large Pixel Detector one can optimize the position of the liquid jet in the X-ray beam for maximal intensity of elastic scattering. The maximal intensity achieved with a pink beam was $3.8 \cdot 10^{15}$. The X-ray emission spectrometer was adjusted by setting the Bragg angle range to ... and orienting the detector perpendicular to the incoming dispersive spectrum (see Figure 3).

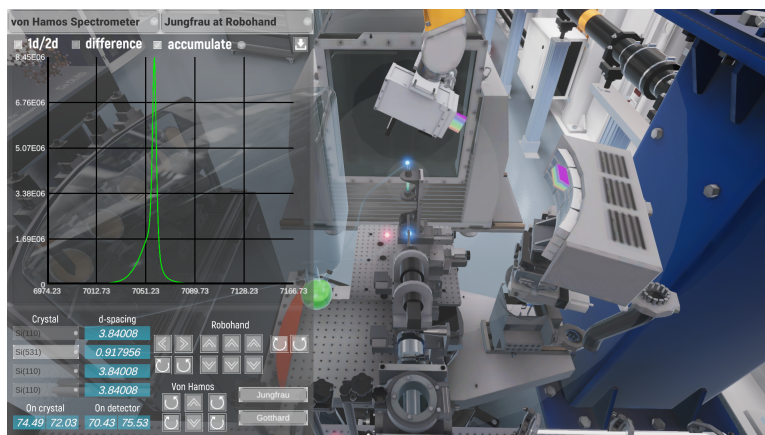


Figure 3: The Von Hamos spectrometer in operating position. with the read-out of the Jungfrau detector as well as the settings for the position to the left.

The excitation laser was set to $\lambda = 400$ nm and $f = 0.02$ with a beam diameter of $d = 150$ μ m in the sample plane. The spatial overlap between laser and X-ray beam was adjusted so that the X-ray laser always overlapped with the optical laser even with the jitter of the optical laser.

First the time delay of the laser was set to 100 ps. This time delay means the laser pulse definitely strikes the sample before the X-ray pulse arrives. XES spectra with the laser on and off were recorded, as well as a difference spectrum.

For the main measurement of the photophysical/photochemical reaction of $[\text{Fe}(\text{bipy})_3]^{2+}$ the range of delays was chosen to be -150 fs to 800 fs with a step size of 50 fs between -150 fs and 400 fs and a step size of 200 fs between 400 fs and 800 fs. For each delay ten difference spectra were recorded with an accumulation time of $60 \text{ s} \pm 2 \text{ s}$ each, which was tracked using a stopwatch.

5. Analysis and Interpretation

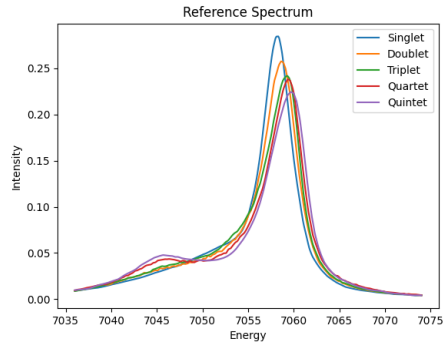


Figure 4: The reference spectra plotted with provided data for different spin states.

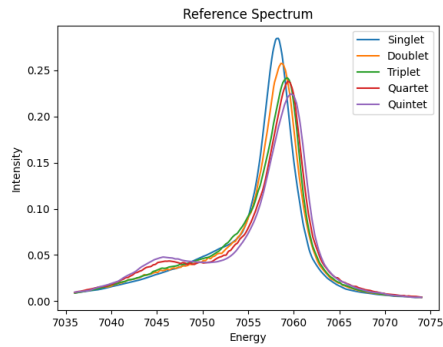
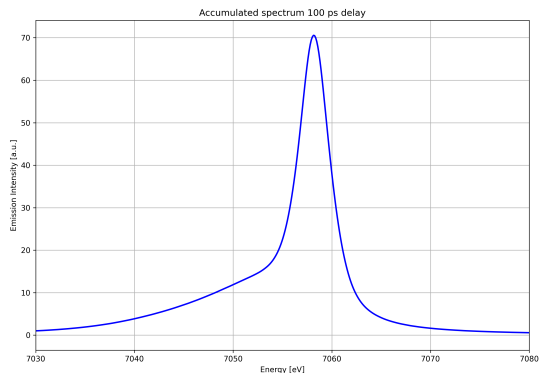


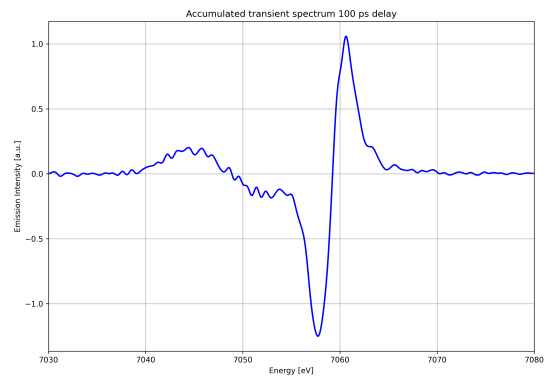
Figure 5: The reference difference spectra with transient signals plotted with provided data for different transitions.

5.1. XES spectra at 100ps

When plotting the XES spectrum and the difference spectrum for a time delay of 100 ps (see Figure ??), a clear transient signal is recognisable with the expected peaks from the reference spectrum (see Figure 5). The XES spectrum shows similarities with the singlet spectrum from the reference (see Figure 4).



(a) XES spectrum with the excitation laser on at a delay of 100 ps.



(b) Difference spectrum at a delay of 100 ps.

6. Discussion of the physical meaning

7. Conclusion and Outlook

References

- [1] P. Willmott, *An Introduction to Synchrotron Radiation*. Wiley, 2019.
- [2] A. Owens, “Synchrotron light sources and radiation detector metrology,” *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, vol. 695, pp. 1–12, 2012.
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